

Discriminant	Number and type(s) of solution(s)
$b^2 - 4ac < 0$	
$b^2 - 4ac = 0$	
$b^2 - 4ac > 0$	

Find the discriminant of the equation and state the number of solutions. · $6p^2 - 2p - 3 = 0$	Find the discriminant of the equation and state the number of solutions. · $-2x^2 - x - 1 = 0$
Find the discriminant of the equation and state the number of solutions. · $-2x^2 - 8x - 14 = -6$	Find all values of m for which the equation has two real solutions. $2t^2 + 8t = -(m+2)$ Reduce all fractions to simplest form.

4.2.6 Notes- Converting Quadratic Standard Form to Vertex Form

Converting Standard Form to Vertex Form

Process

Convert standard form to vertex form by completing the square.

1. Move the constant term to the left side.
2. Remove a numeric factor from the quadratic and linear term if the quadratic term has a coefficient other than 1.
3. Add a value to each side that makes the quadratic expression a perfect square trinomial. (*Remember to multiply by the numeric factor on the constant term side!*)
4. Factor the perfect square trinomial.
5. Simplify the constant terms and move them to the same side as the factored quadratic expression.

$$f(x) = 3x^2 - 18x + 17$$

$$f(x) - 17 = 3x^2 - 18x$$

$$f(x) - 17 = 3(x^2 - 6x)$$

$$f(x) - 17 + \boxed{} = 3(x^2 - 6x + \boxed{})$$

$$f(x) - 17 + \boxed{3(9)} = 3(x^2 - 6x + \boxed{9})$$

$$f(x) - 17 + 27 = 3(x - 3)^2$$

$$f(x) + 10 = 3(x - 3)^2$$

$$f(x) = 3(x - 3)^2 - 10$$

Hint: Use $\left(\frac{b}{2a}\right)^2$ to calculate the "box" number for the trinomial.

Example 1: Convert the following into vertex form:

a) $f(x) = -7x^2 + 42x - 54$

b) $f(x) = 2x^2 - 24x + 65$